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ELECTROWETTING FOR MULTI-SCALE DROPLET COALESCENCE AND TRANSPORT IN DROPWISE CONDENSATION

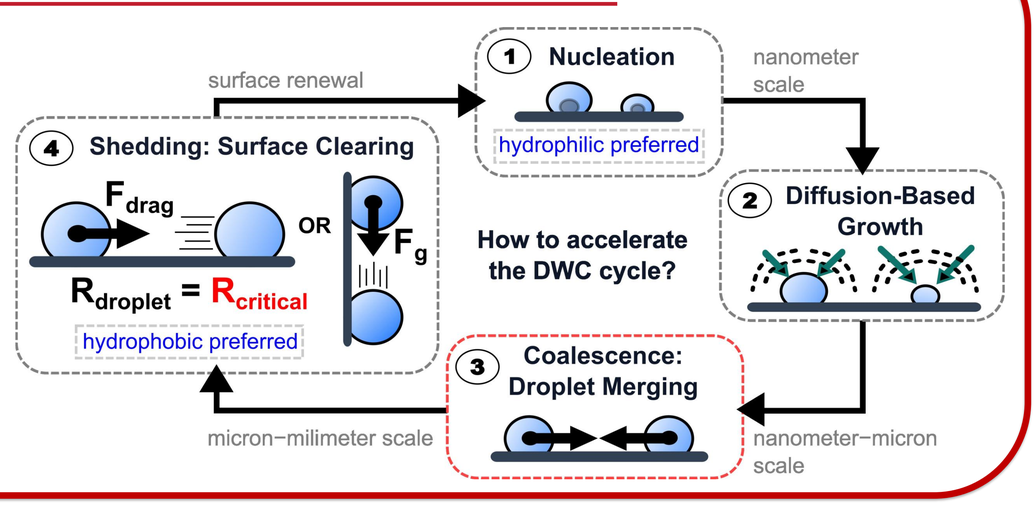
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INTRODUCTION & MOTIVATION

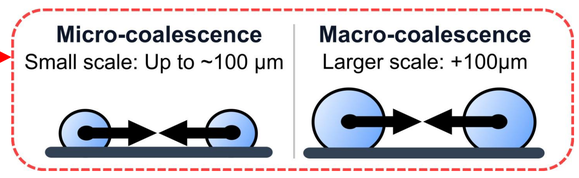
- Condensation central to water harvesting, air conditioning, and power generation.
- **Dropwise condensation** (DWC) is preferred over filmwise for greater heat transfer (up to ~10× better).
- DWC is cyclical: shorter cycle → higher heat transfer.
- **Current bottlenecks**:

(i) Conflicting wettability requirements
(ii) Gradual coalescence towards the critical shedding size ($R_{critical}$)



DWC ACCELERATION STRATEGY

- **Objectives**: (i) Improve *micro-coalescence* and (ii) promote *macro-coalescence* to ripen droplets to critical size.



Enabler

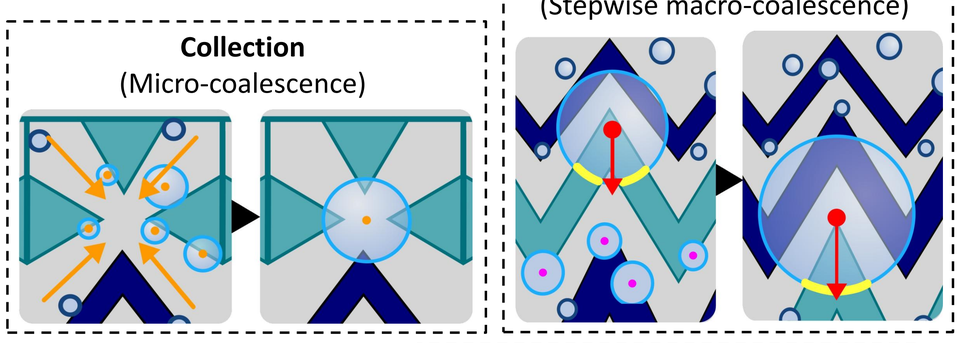
Electrowetting (EW) uses electric fields to modify surface energy.

- EW offers wettability control (θ_V) on application of voltage ($V_{applied}$) → droplet contact line oscillation via alternating current signal.

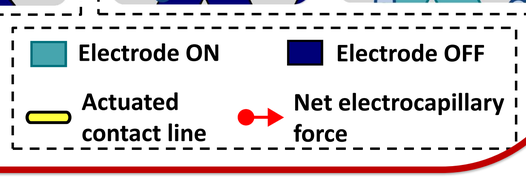
$$\left(\cos(\theta_V) = cos(\theta_0) + \frac{\varepsilon_0 \varepsilon}{2dy} \boldsymbol{V^2_{applied}}\right)$$

EW enables our two-part collection & transport mechanism:

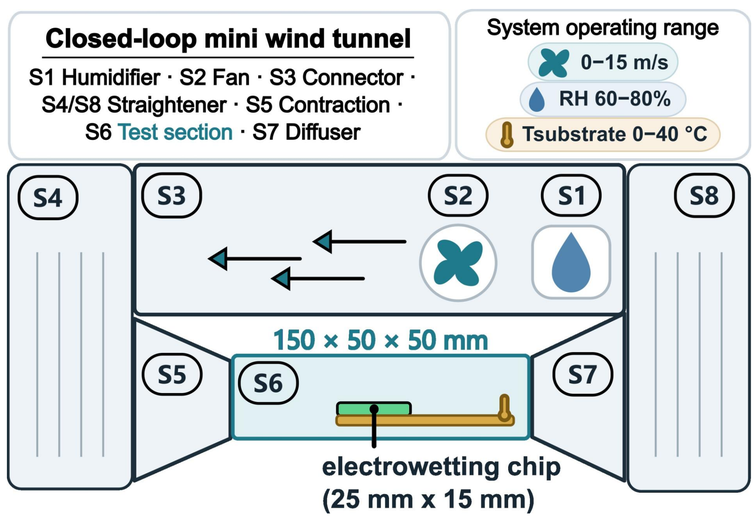
- Collection concentrates *micro-coalescence*
- Transport directs droplets away from collection zone, coarsening them via stepwise control



- **How?** Asymmetric electrode geometry & arrangement → electric fields produce electrostatic energy gradient → net electrocapillary force → directed droplet motion (eventual shedding at R_{cr} by drag force)

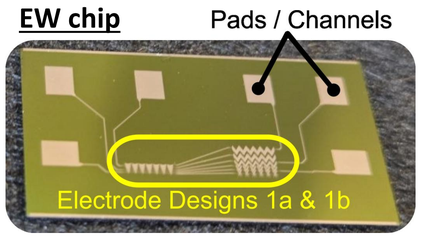


EXPERIMENTAL SETUP



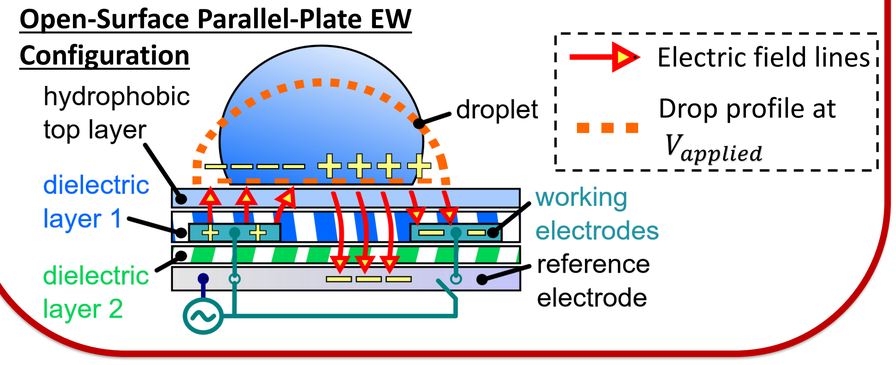
- A horizontal condensation environment was simulated using a closed-loop mini-wind tunnel.
- **Conditions**: RH: 60% , Air velocity: 1 m/s, Tsubstrate: 5°C, Signal: 120 V peak-to-peak (Vpp), 250 Hz

EW chip



- A GUI-controlled circuit activates selected electrode channels while an overhead high-speed camera recorded droplet growth and transport.

Open-Surface Parallel-Plate EW Configuration



RESULTS

Collection:

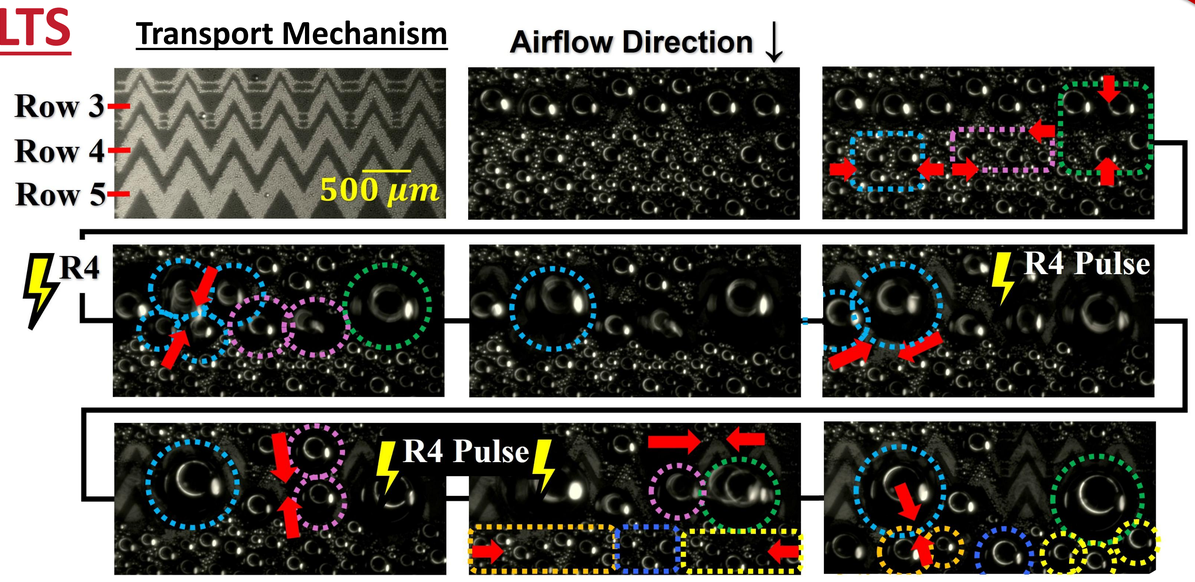
- Row 1 activation doubled droplet diameter from ~20–40 μm in ~5 s; inactive rows remained < 20 μm.
- Coarsening remained localized to active electrode edges, consistent with repeated micro-coalescence.



Transport:

- Sequential activation / pulsing of rows induced directed motion and repeated macro-coalescence
- Droplet diameters coarsened from ~250-350 μm to ~800 μm diameter in ~30 s

Transport Mechanism



- Upstream rows were cleared on transport, re-exposing surface for renewed nucleation.

COMPARATIVE ANALYSIS

Study	T_{air} (°C) / RH (%)	T_{surf} (°C)	ΔT_{dew} (°C)	Δp_v (kPa)
Baratian et al. [1]	36 / 100	25	~ 11	~ 2.8
Hoek et al. [2]	42 / 99	11.5	~ 30	~ 6.9
Bonab et al. [3]	25 / 60	0	~ 17	~ 1.3
Current study	**25 / 60**	**5**	**~ 12**	**~ 1.0**

Study	Time to ~ 800 μm (s)	U (m/s)	V_{pp} (V)	f (Hz)
Baratian et al. [1]	~ 900	–	~ 424	1000
Hoek et al. [2]	~ 570	–	~ 424	1000
Bonab et al. [3]	~ 900	1	~ 424	10000
Current study	**~ 46**	**1**	**120**	**250**

- Our ~800 μm was reached at the lowest condensation potential among the studies compared, thus faster coarsening can be largely attributed as product of our acceleration mechanism.
- The dual-mechanism and configuration delivered rapid surface renewal and strong actuation at 120 Vpp. In [3], ~800 μm in ~46s matched closely to the current work, but at 15 m/s and 424 Vpp.
- Despite lower condensation potential and applied voltage, the ~46 s coarsening time was roughly one order of magnitude shorter than prior reports of ~570–900s.

CONCLUSIONS

- Micro-coalescence and macro-coalescence can promote **rapid multiscale droplet coarsening** to shedding sizes and renew the upstream surface.
- **~800 μm diameters in ~46 s** at the **lowest condensation potential**, roughly an order of magnitude shorter than comparable conditions in prior works.
- Shedding and HTC measurements are needed to confirm thermal enhancement.

REFERENCES

[1] Baratian et al., *Phys. Rev. Lett.* 120, 214502 (2018)
[2] Hoek et al., *Adv. Mater. Interfaces* 8, 2001317 (2021)
[3] Bonab et al., *Int. J. Heat Mass Transf.* 220, 124925 (2024) · Full paper: IHTC18-0611 · Funding: NSERC

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